Zero Inflated Count Regression Article

1 Background

1.1 Zero Inflated Count Regression

Zero inflated count regression is a modification of a generalized linear model following a count distribution, like Poisson or Negative Binomial. When our data has more counted zeros than the model would predict, this usually causes the model to be overdispersed, and the effects of terms in the model to be *overestimated*. To adjust for this, we treat the zeros as being predicted by a logistic regression model, while all the points greater than zero are predicted by the specified count regression model.

Where a typical negative binomial regression has the form:

$$log(\mu_{Y_i|x_i}) = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_{p-1} x_{(p-1)i}$$

where $\mu_{Y_i|x_i}$ is the mean of the Negative Binomial distribution from which the ith response is drawn.

A zero inflated negative binomial model predicts the zeros from a logistic model:

$$log\left(\frac{\pi_i}{1-\pi_i}\right) = \alpha_0 + \alpha_1 x_{1i} + ... + \alpha_{p-1} x_{(p-1)i}$$

where $log\left(\frac{\pi_i}{1-\pi_i}\right)$ is the log odds of a structural 0.

Also, two models don't need to have the same set of predictor x_i's.

Note: The same transformation can be applied to poisson models as well as models with an offset.

We usually care about interpreting the negative binomial model more than the logistic model, but what causes zeros can also be interesting.

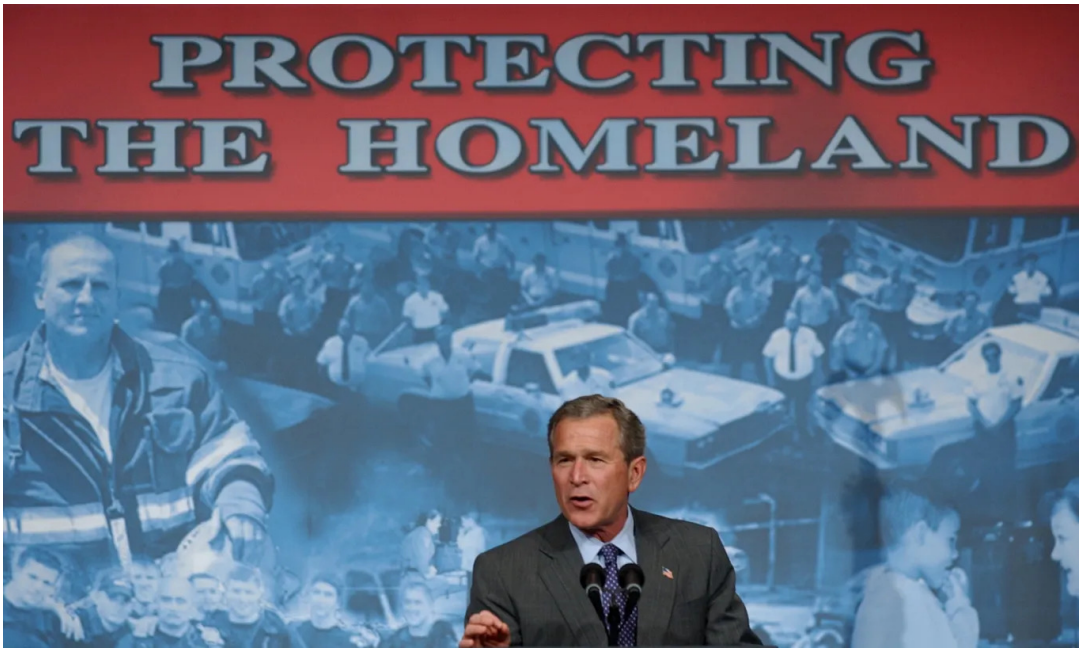
1.2 The Study

Figure 1: Nick Anderson



Surveillance, in any form, has long been controversial and politicized. Foucalt used the example of the panopticon to illustrate the way constant surveillance suffocates freedom and assures the maintenance of power (Foucault 2012). 25 Years ago, the war on terror in the United States served as the pretense for the passing of the Patriot Act, allowing the U.S. government greater access to personal digital information for the ostensible cause of anti-terrorism (King 2026).

Figure 2: Newsweek



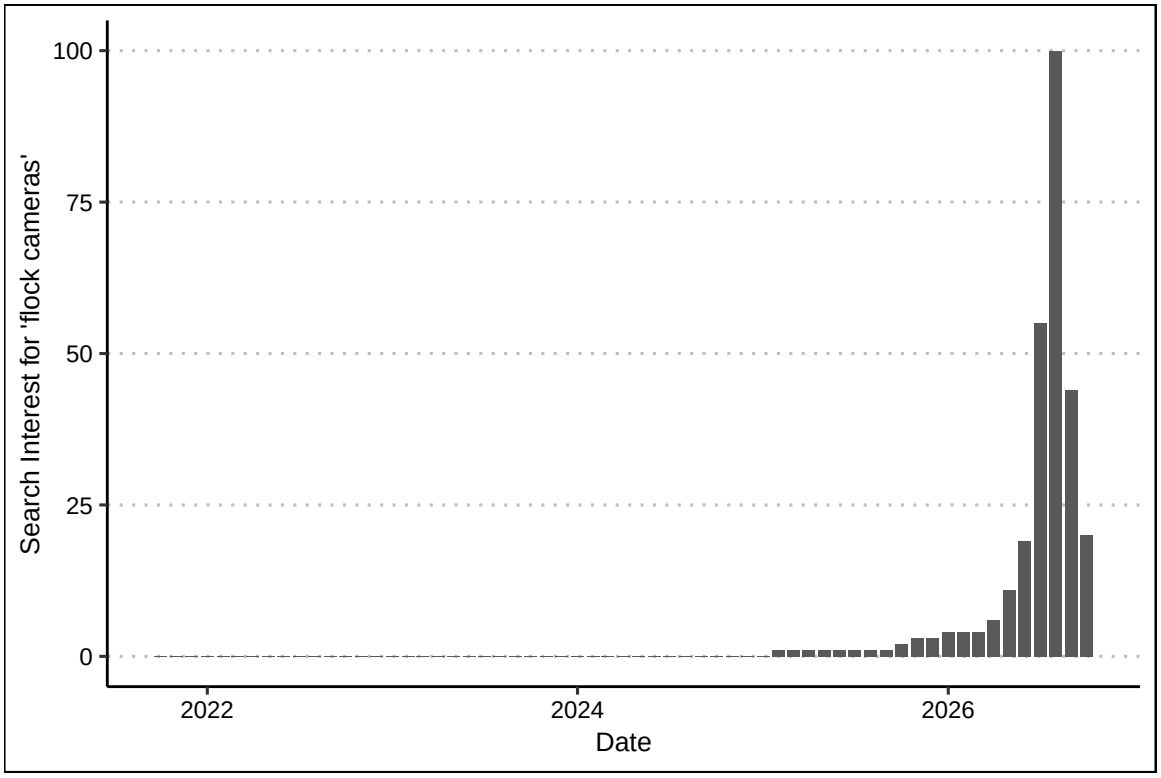
Where surveillance in previous decades was limited by an understanding that the guard in the panopticon tower could not be constantly watching every person, and a government official could not sift through every possible online document, the ability of large language models to scrape information from massive databases in real time has pushed the envelope for surveillance.

Figure 3: WBAL



Security camera adoption has exploded in recent years, as algorithmic video detection models, like those used in cameras manufactured by Flock Safety, has led to data collection and surveillance on an unparalleled level (Brumbaugh 2026).

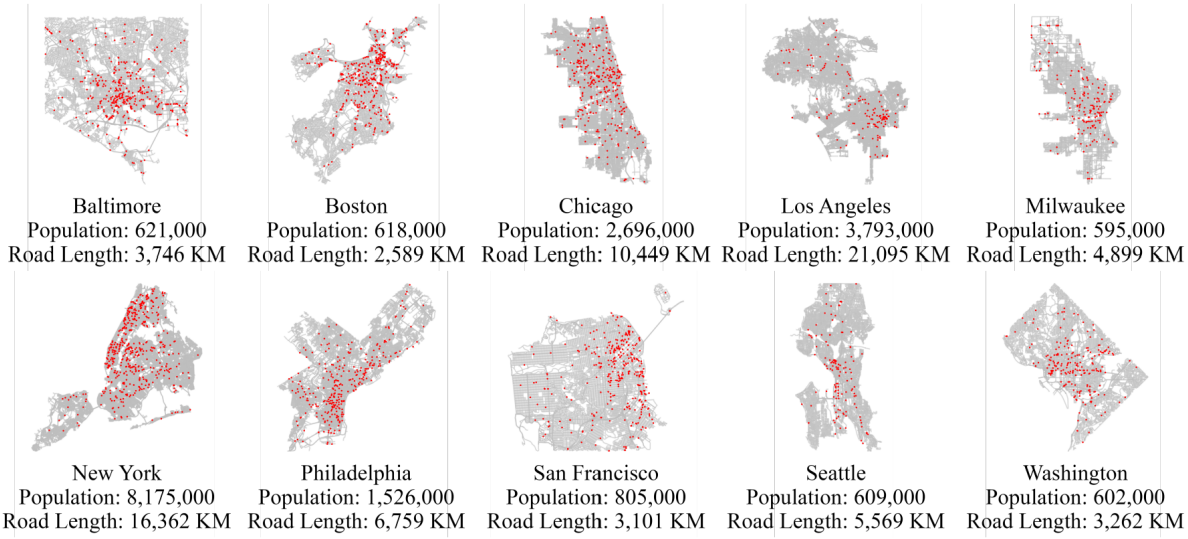
Figure 4: Google Trends of Flock Cameras



The article by Dahir et al. titled, "Surveillance camera prevalence and racial diversity in ten US cities", published in 2025, looks to see how the prevalence and density of surveillance cameras in major U.S. cities differs by demographic and socioeconomic characteristics of the areas they're in. They look to investigate if surveillance cameras are highest in homogeneous black neighbrhoods, as that would confirm the historical theory that black people often live in neighborhoods with characteristics resembling the carceral system. Another hypothesis says that surveillance cameras might be highest in more diverse neighborhoods where white residents might be more inclined to use the cameras for perceived safety and control. For example, white people who might be perceived as 'gentrifiers' in historically diverse neighborhoods might take pride in heterogeneous demographics while still feeling a sense of comfort with the added security of a camera (Dahir et al. 2025).

To collect their count of surveillance cameras, the authors first trained a machine learning image detection model on confirmed camera locations the ten most dense populated cities in the United States:

Figure 5: Dahir et al.



After model training, they randomly selected 100,000 points within each city, went to each of those points with Google Street View, and captured a 360° photo of the surroundings. They they ran all those photos through the camera detection models, and had humans validate the existence of cameras in photos flagged by the model (those poor undergrads). The researchers estimate that around 60% of cameras in the photos were detected by the model, so the magnitude of the data is likely undercounted. The count of the cameras was then combined with publicly available demographic data from where each photo was taken to create a combined dataset.

Figure 6: Dahir et al.



1.3 Goal of Statistical Analysis

Because of the countable nature of the data, the researchers propose using a generalized linear model with a poisson distribution. Because of a large amount of images with no cameras taken, it makes sense to include the zeros in the model structurally with a logistic component.

I think their main analysis points are seeing how camera amounts change with regards to the white %, black%, and overall diversity of the areas, while accounting for other socioeconomic and strucutural confounding variables like income levels, crime rate, zoning, etc.

1.3.1 Formula Antics

I had a somewhat hard time determining the correct formula and terms for the regression equation. Below I'll illustrate several reasons why:

In the methods, the authors say:

> " We also expect excess zeros to be generated by a separate process from counts."

I didn't immediately know what this meant without looking at the code. After finding the equation I assume they used by looking in the code, I found this quote which is, in a choice I find puzzling, separate from the quote above:

> "The zero component accounts for the city and road length in all models."

Which, by confirming with their code, means the logistic element of their zero-inflated model is only regressed on the city with the road length as an offset. However, I don't understand why the other metrics in the model wouldn't also contribute to the detection of zero cameras, so I think it makes sense to leave them in both sides of the model.

They mention the inclusion of quadratic terms for the racial demographic variables (in the results but not the methods):

> "Given the parabolic relationships observed in the bivariate models, this model includes quadratic terms for both Black and white residents."

I take this to mean they adjusted the model after seeing a quadratic looking relationship in some of their effects in a "bivariate model". I can only assume this because there's no mention of a "bivariate model" in their methods, so the reader is left to wonder if that was the original hypothesis, a planned arm of their analysis, or something else. Still, the inclusion of the quadratic terms makes sense as the demographic variables are percentages and not truly continuous, however I'm not sure why they don't mention this explicitly in the methods. However, if these terms were only added because of a noticed trend in their data, this is basically 'HARKing', or "hypothesizing after the results are known," which is generally frowned upon in research (Taylor 2025).

Lastly, the authors mention the use of images taken as a weight component in the model:

> "We include … a weight for the image count."

This weights the data points with more images taken more heavily in the model. I think this highlights a potential issue with the data collection, and regardless of the choice of including or not including the weights, each has its own issues. If we don't include the images taken as a weight, a single image taken for one observation is weighted the same as an observation with over a hundred images taken; clearly one of those is more comprehensive. However, not all images are created equal, and images taken in close proximity to each other don't necessarily add importance to that observation. Because of this, I'll go forward with both options, seeing if the weights significantly alter the model.

With all that being said here's the regression equation I personally think makes sense:

Predictors (Poisson Counts and Logistic Zeros):

- % White (quadratic and linear)
- % Black (quadratic and linear)
- Overall Racial Diversity
- Crime Rate
- Population
- Median Household Income (in thousands)
- Home Vacancy Rate
- Median Home Value (in thousands)

- Neighborhood Designation (residential, public, commercial, industrial, roads, or mixed)
- City (Baltimore, Boston, Chicago, Los Angeles, Milwaukee, New York City, Philadelphia, San Francisco, Seattle, or Washington D.C.)

Offset:

- Road length (km)

Weights (potentially):

- log(images taken)

In *R*, our formula call looks like this:

```
formula <- cameras ~ white + I(white^2) + black + I(black^2) + diversity + crime +
    pop + income + vacancy + home_value + zone + city
```

2 Reproduction of the Study

2.1 Data Cleaning

I mainly renamed the variables to my liking, as well as scaling the continous metrics because that made the model behave better later on (not shown).

```
dat <- read_csv('cameras.csv')

dat <- dat |>
  rename(cameras = cam_count,
         images = image_count,
         white = pnhwht,
         black = pnhblk,
         diversity = entropy,
         crime = total_crime_rate,
         income = hinc,
         vacancy = pvac,
         home_value = mhmval,
         zone = modal_zone,
         road_length = road_length_km)

dat$diversity <- as.vector(scale(dat$diversity))
```

2.2 Data Summary

Let's see how the data is split up by city:

Our dataset is heavily influenced by New York City. We also see that most of the cities are relatively close in diversity, even with large differences in percentage of white and black populations. Lastly, there's not much we can get from the cameras because there's so many zeros.

Here's a rough correlation analysis between white%, black%, diversity% and cameras:

Hard to see any pattern in the demographic factors against the camera counts.

2.3 Model Assumptions

2.3.1 Zero Inflation

First let's see, how is zero inflated is the model compared to what we'd expect under a standard poisson regression:

We never even come close to simulating as many zeros as there are observed. This definitely warrants a zero inflated model.

Figure 7: Pairwise Summary of Data

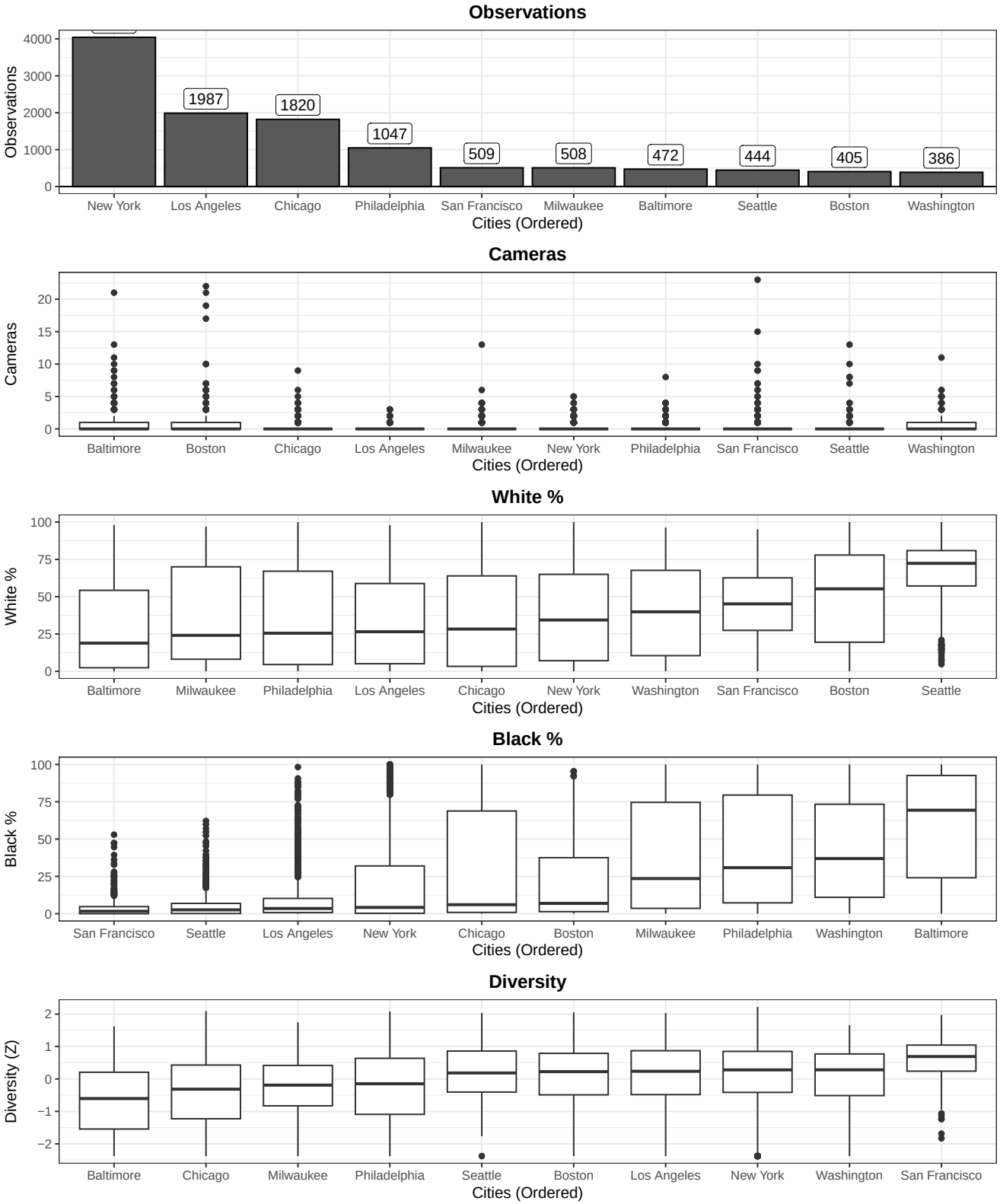


Figure 8: Correlation Analysis

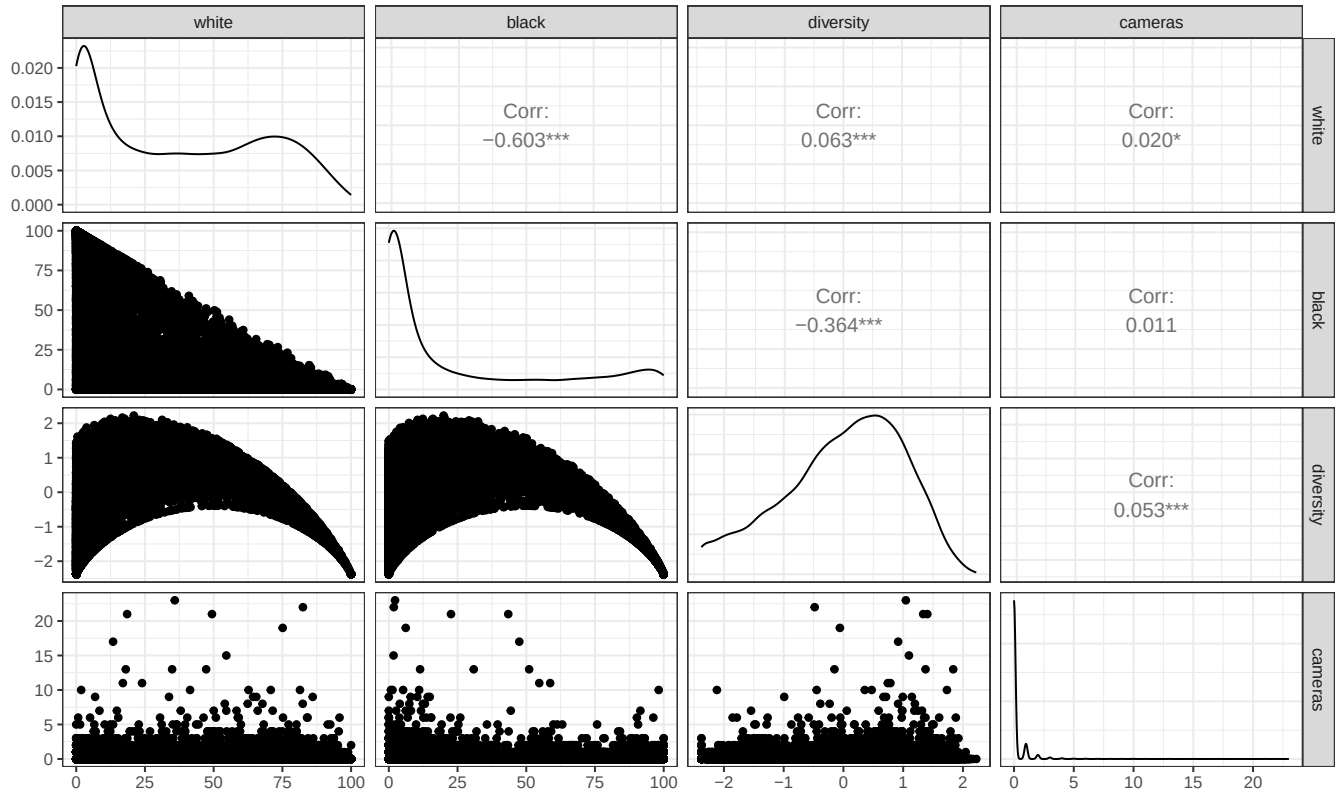
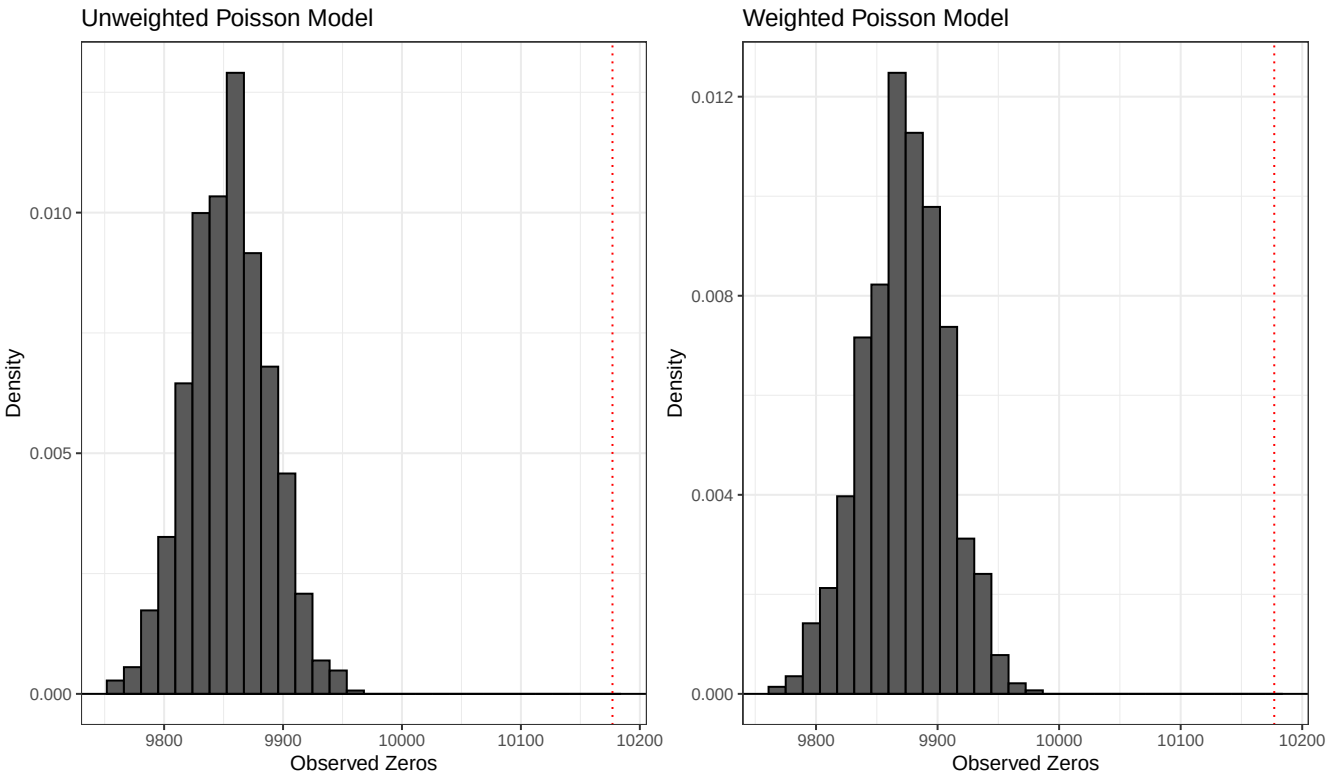


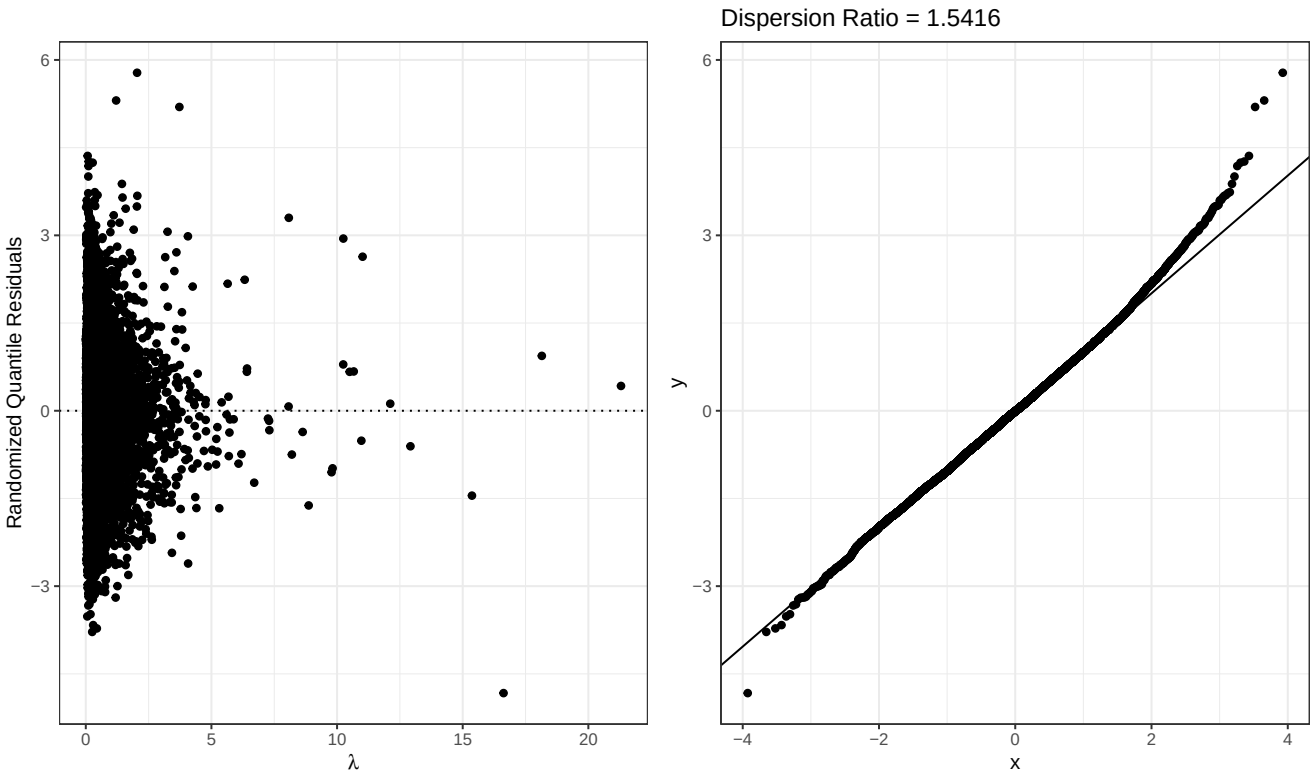
Figure 9: Zero Inflation Testing of the Standard Poisson Model



2.3.2 Dispersion

Here we'll test the randomized quantile residuals of the zero inflated model and the dispersion ratio. First, the unweighted model:

Figure 10: Randomized Quantile Residuals (Unweighted Model)



The randomized quantile residuals look pretty good, as there's a relatively even band of points. The dispersion ratio is a tad high but still not bad. There's a few outliers but that's to be expected with a very large sample size.

Next, the weighted model:

Adding the weights significantly increases the overdispersion of the model.

We can also check our predictions against the standardized Pearson residuals:

Looking more closely around 1, we see an increasing curved pattern as λ increases. This might suggest that a negative binomial might be better suited, but combined with the dispersion ratio and randomized quantile residuals, I think a poisson model is pretty well justified here.

The curved pattern seen before is even worse for the weighted model.

2.3.3 Multicollinearity

Because the design matrix is the same for both models, the VIF for both models is:

Table 1: Model VIF

Term	VIF	Lower CI	Upper CI	Model Component
White %	4.772	4.643	4.883	conditional
(White %)^2	3.938	3.833	4.029	conditional
Black %	8.235	8.009	8.429	conditional
(Black %)^2	10.575	10.285	10.826	conditional
Diversity	7.794	7.581	7.978	conditional

Table 1: Model VIF

Term	VIF	Lower CI	Upper CI	Model Component
Crime Rate	1.487	1.456	1.513	conditional
Population	1.425	1.397	1.449	conditional
Median Income	3.464	3.372	3.543	conditional
Vacancy Rate	1.368	1.342	1.390	conditional
Median Home Value	4.041	3.933	4.134	conditional
Zoning	23.407	22.770	23.972	conditional
City	100.364	97.664	102.821	conditional
White %	5.093	4.955	5.211	zero inflated
(White %)^2	4.312	4.197	4.412	zero inflated
Black %	7.548	7.342	7.726	zero inflated
(Black %)^2	9.524	9.263	9.749	zero inflated
Diversity	7.596	7.388	7.775	zero inflated
Crime Rate	1.624	1.589	1.655	zero inflated
Population	1.147	1.131	1.160	zero inflated
Median Income	2.721	2.651	2.781	zero inflated
Vacancy Rate	1.254	1.232	1.271	zero inflated
Median Home Value	3.407	3.317	3.485	zero inflated
Zoning	7.648	7.442	7.831	zero inflated
City	41.595	40.476	42.613	zero inflated

To be expected, a lot of the variables have very high correlations with other variables, especially the demographic characteristics of each city. While I'll keep the city term in the model to compare with the paper's analysis, I do think it's probably not providing much necessary information that's not contained in other variables.

2.3.4 Sufficient Sample Size for Predictors

Ideally we should have at least 10 events counted per predictor:

$$\sum_{i=1}^{n=11,620} \lambda_i = 2524 \geq 10p = 10(25) = 250 \quad \checkmark$$

2.4 Model Fit

We can obtain an overall model fit of both models. First, the unweighted model:

Table 2: Model Fit

AIC	AICc	BIC	R2	R2_adjusted	RMSE	Sigma	Score_log	Score_spherical
10470.73	10471.17	10838.75	0.186	0.184	0.67	0.671	-0.446	0.009

Our unweighted model explains 18.4% of the variance in surveillance cameras.

Now the weighed model:

Table 3: Model Fit

AIC	AICc	BIC	R2	R2_adjusted	RMSE	Sigma	Score_log	Score_spherical
44517.89	44518.33	44885.91	0.062	0.06	1.522	3.598	-0.447	0.009

Our weighted model explains 6% of the variance in surveillance cameras.

Figure 11: Randomized Quantile Residuals (Weighted Model)

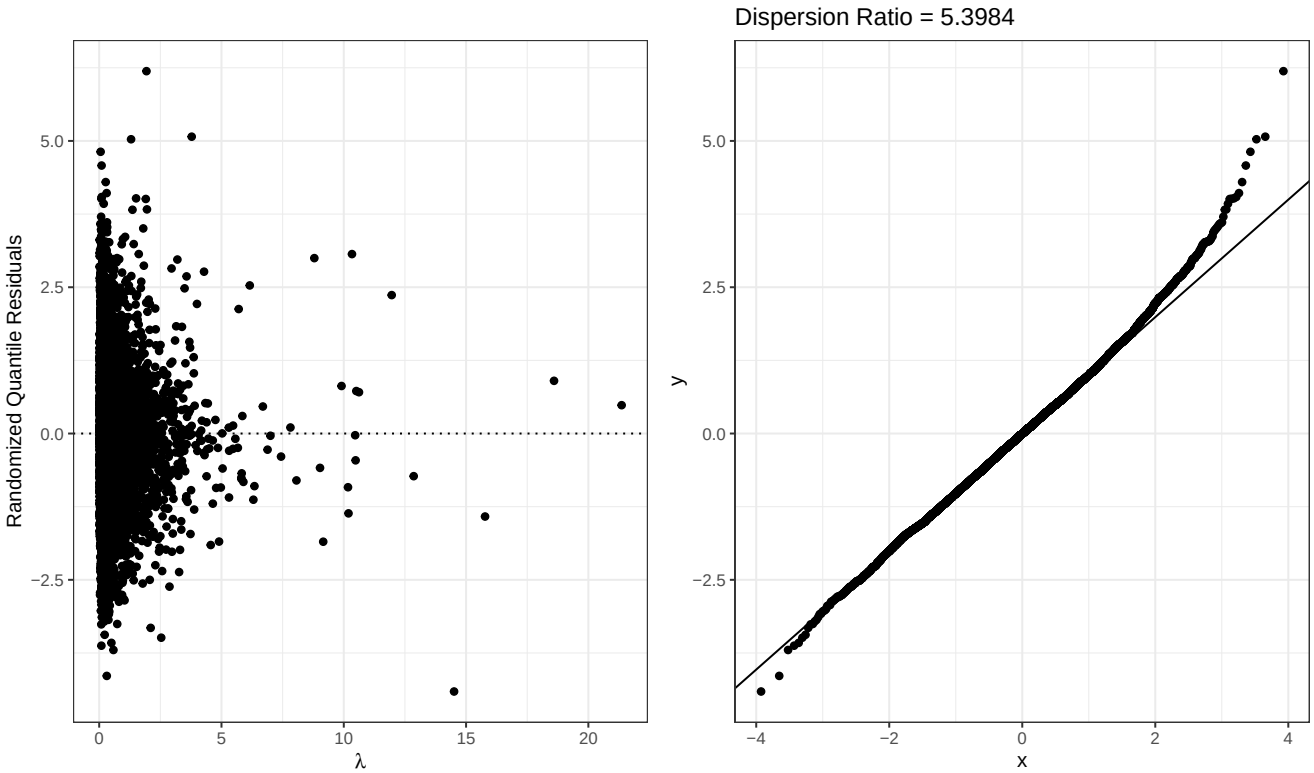


Figure 12: Predictions Against Pearson Residuals (Unweighted Model)

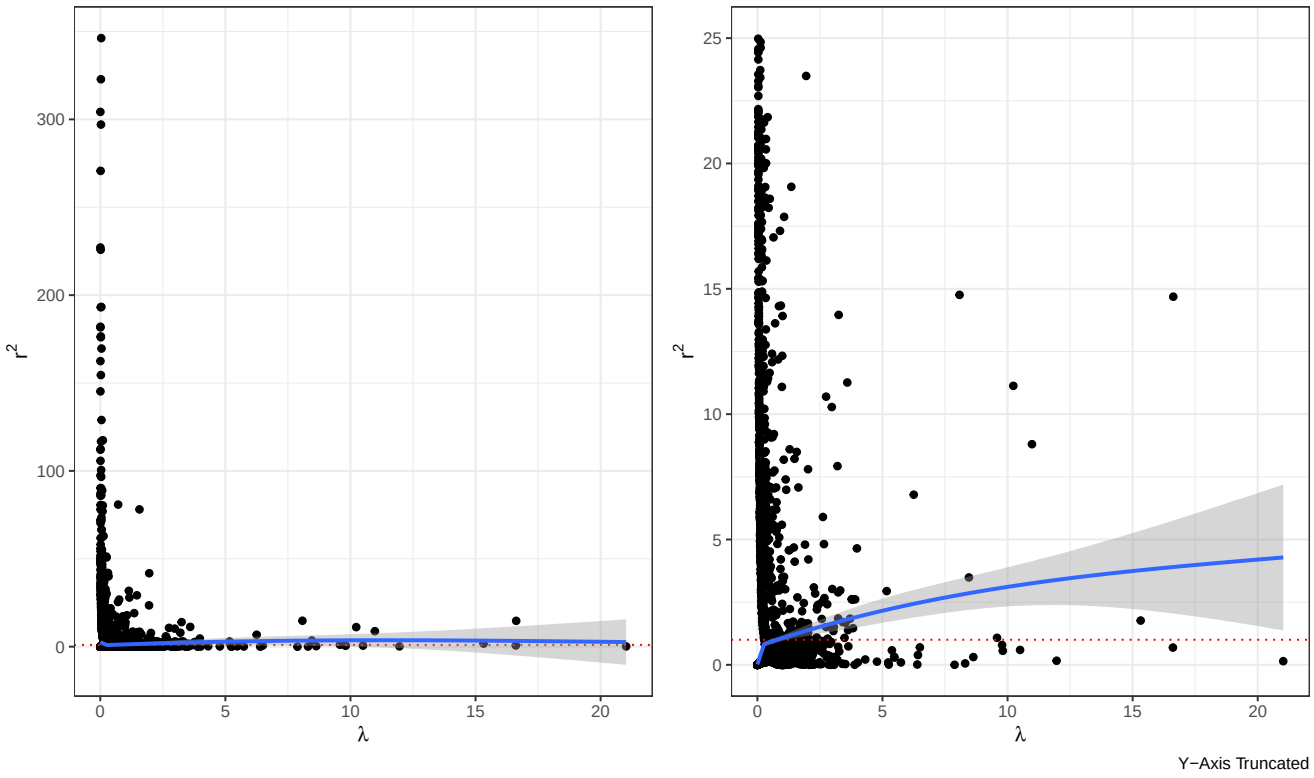
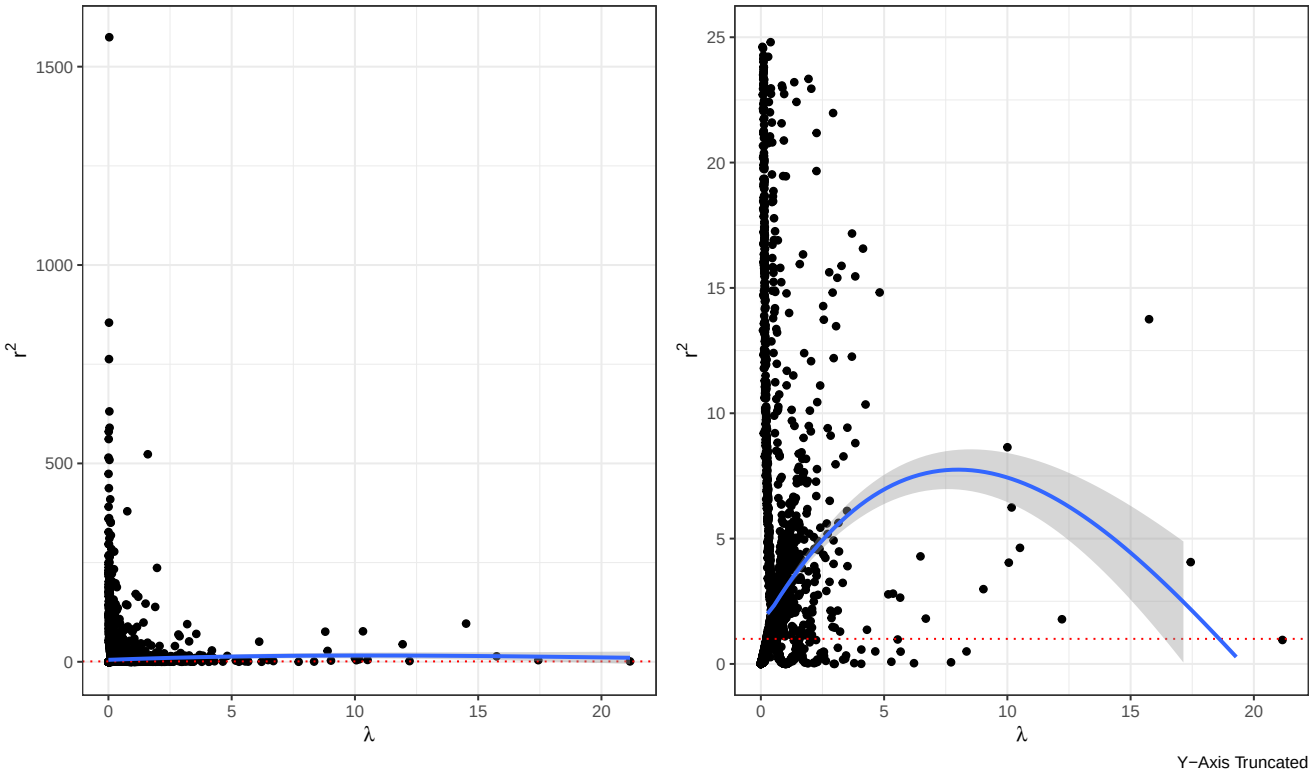


Figure 13: Predictions Against Pearson Residuals (Weighted Model)



Remember that some of the model fit statistics are non-comparable because they're calculated differently for weighted and non-weighted models.

We can directly compare the model fits with the cross-validated root mean square error:

This tells us, that by their ability to predict, the models are almost identical.

2.5 Model Inference

Now to the heart of the analysis:

- How many surveillance cameras are in highly white or highly black neighborhoods?
- How many surveilance cameras are in more diverse neighborhoods?
- Are there other confounding factors that affect surveillance cameras?

A general summary our model gives us:

where the percent change = $(e^{\text{estimate}} - 1) \times 100$.

Testing the effect sizes of our fixed effects (**r2glmm?**):

2.6 Model Interactions

2.7 Conclusions

3 Disagreements With the Article

4 Acknowledgement

I would like to thank the researchers who published this paper: Shihong Jia, Xugao Wang, Zuoqiang Yuan, Fei Lin, Ji Ye, Guigang Lin, Zhanqing Hao & Robert Bagchi. Transparency in data and code availability is hugely important

for research ethics, and it makes my project possible.

5 AI Use

All of the guidance given by generative AI for this article can be found in this conversation.

6 Statistical Packages Used

- *tidyverse* for… everything it's *tidyverse* (Wickham et al. 2019)
- *ggplot2* for plotting (Wickham 2016)
- *ggthemes* for theme_clean() (Arnold 2024)
- *patchwork* for combined plots (Pedersen 2025)
- *pscl* for inference of Zero Inflated Count Regression model (Zeileis et al. 2008)
- *performance* for model fit statistics (Lüdecke et al. 2021)
- *emmeans* for interaction analysis (Lenth 2025)

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